Splitting Cell Into Multiple Rows In Pandas DataFrame

Python

January 28, 2025 Evans Samantha

Pandas is a powerful data manipulation library in Python that provides various functions and methods to work with structured data. One common task when working with data is splitting a cell that contains multiple values into multiple rows in a DataFrame. This article will explore different approaches to achieve this in Pandas.

Approach 1: Using the str.split() method

The str.split() method in Pandas can be used to split a string into multiple values based on a delimiter. To split a cell into multiple rows, we can follow these steps:

```
import pandas as pd

# Create a sample DataFrame
data = {'Name': ['John, Doe', 'Jane, Smith'],
        'Age': [25, 30]}
df = pd.DataFrame(data)

# Split the 'Name' column into multiple rows
df['Name'] = df['Name'].str.split(',')

# Explode the 'Name' column into multiple rows
df = df.explode('Name')
```

In the above code, we first create a sample DataFrame with a 'Name' column that contains multiple values separated by a comma. We then use the str.split() method to split each cell in the 'Name' column into a list of values. Finally, we use the explode() function to explode the 'Name' column into multiple rows, creating a new row for each value in the list.

Approach 2: Using the melt() function

The melt() function in Pandas can also be used to split a cell into multiple rows. The melt() function unpivots a DataFrame from wide format to long format, allowing us to specify the columns to keep as identifiers while melting the rest into a single column. Here's an example:

```
import pandas as pd

# Create a sample DataFrame
data = {'Name': ['John, Doe', 'Jane, Smith'],
        'Age': [25, 30]}
df = pd.DataFrame(data)

# Split the 'Name' column into multiple rows using melt()
df = df.melt(id_vars='Age', value_name='Name').drop('variable', axis=1)
df['Name'] = df['Name'].str.split(',')

# Explode the 'Name' column into multiple rows
df = df.explode('Name')
```

In this approach, we first use the melt() function to unpivot the DataFrame, keeping the 'Age' column as an identifier and melting the 'Name' column into a single column called 'Name'. We then split each cell in the 'Name' column into a list of values using the str.split() method and explode the 'Name' column into multiple rows.

Approach 3: Using the apply() function

Another approach to split a cell into multiple rows is by using the apply() function along with a lambda function. Here's an example:

```
import pandas as pd

# Create a sample DataFrame
data = {'Name': ['John, Doe', 'Jane, Smith'],
        'Age': [25, 30]}
df = pd.DataFrame(data)

# Split the 'Name' column into multiple rows using apply()
df['Name'] = df['Name'].apply(lambda x: x.split(','))

# Explode the 'Name' column into multiple rows
df = df.explode('Name')
```

In this approach, we use the apply() function to apply a lambda function to each cell in the 'Name' column. The lambda function splits the cell into a list of values using the str.split() method. Finally, we explode the 'Name' column into multiple rows using the explode() function.

Splitting a cell into multiple rows in a Pandas DataFrame can be achieved using various approaches. The str.split() method, melt() function, and apply() function are some of the methods that can be used to accomplish this task. By understanding and applying these techniques, you can efficiently manipulate and transform your data in Pandas.

Here are some examples related to splitting a cell into multiple rows in a Pandas DataFrame:

Example 1:

```
import pandas as pd

# Create a DataFrame
data = {'Name': ['John', 'Alice', 'Bob'],
        'Skills': ['Python, Java', 'C++, Python', 'Java']}
df = pd.DataFrame(data)

# Split the 'Skills' column into multiple rows
df = df.assign(Skills=df['Skills'].str.split(', ')).explode('Skills')

print(df)
```

This example demonstrates how to split the 'Skills' column into multiple rows using the `str.split()` function and the `explode()` function. The resulting DataFrame will have one row for each skill.

Example 2:

```
import pandas as pd

# Create a DataFrame
data = {'Name': ['John', 'Alice', 'Bob'],
        'Subjects': ['Math, Science', 'English', 'History']}
df = pd.DataFrame(data)

# Split the 'Subjects' column into multiple rows
df = df.assign(Subjects=df['Subjects'].str.split(', ')).explode('Subjects')

print(df)
```

In this example, the 'Subjects' column is split into multiple rows using the same approach as in Example 1. The resulting DataFrame will have one row for each subject.

Example 3:

```
import pandas as pd

# Create a DataFrame
data = {'Name': ['John', 'Alice', 'Bob'],
        'Languages': ['English', 'Spanish, French', 'German']}
df = pd.DataFrame(data)

# Split the 'Languages' column into multiple rows
df = df.assign(Languages=df['Languages'].str.split(', ')).explode('Languages')

print(df)
```

This example demonstrates how to split the 'Languages' column into multiple rows. The resulting DataFrame will have one row for each language.

Reference links:

- Pandas documentation on str.split()
- Pandas documentation on explode()

Conclusion:

Splitting a cell into multiple rows in a Pandas DataFrame is a useful technique when dealing with data that is stored in a delimited format within a single cell. By using the `str.split()` function and the `explode()` function, we can easily split the cell values into separate rows. This allows for better analysis and manipulation of the data. It is important to note that the original DataFrame will be modified, so it is recommended to create a copy if the original data needs to be preserved.

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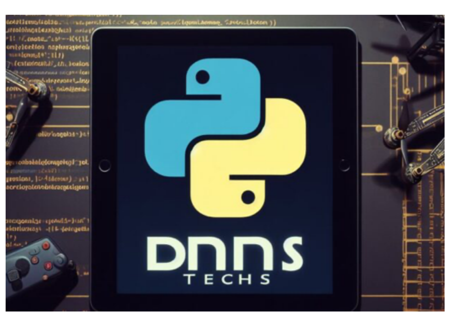

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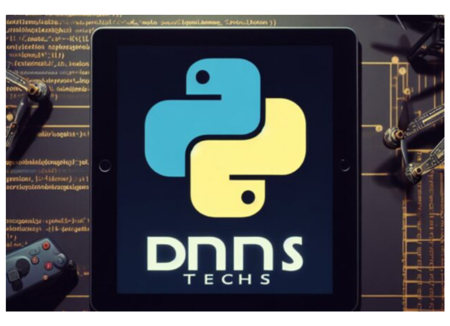

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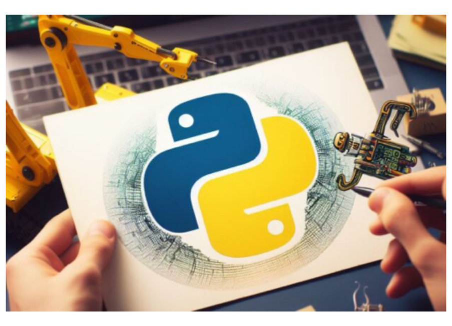

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